

MePe: Rethinking Multimodal Chinese Idiom Reading Comprehension from a Metaphorical Perspective

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Abstract

The multimodal Chinese idiom reading comprehension task aims to select the most appropriate idiom from a candidate list via the given text and image. This poses a significant challenge for the model to comprehend each Chinese idiom accurately. Existing multimodal Chinese idiom reading comprehension methods primarily focus on aligning contextual text and images, while overlooking two key attributes of Chinese idioms. (1) There is a discrepancy between the literal and metaphorical meanings of Chinese idioms. (2) The same Chinese idiom has different meanings in different scenarios, which requires targeted understanding by experts who specialize in different fields. To address the above challenges, we rethink the solution to the multimodal idiom reading comprehension task from a metaphorical perspective and propose a framework named MePe. Firstly, we propose a literal metaphorical semantic graph that systematically transforms the implicit discrepancy between the literal and metaphorical meanings of Chinese idioms into structured explicit relationships, thereby making metaphorical meanings more understandable. Then, we propose a mixture of idiom experts consisting of a literal idiom expert and a metaphorical idiom expert. Through division of labor and collaboration among these experts, we achieve an understanding of the dual meanings of Chinese idioms across different scenarios. Finally, we employ the maximum mean discrepancy to adjust the variance between the literal and metaphorical semantic features of Chinese idioms. By mapping these features onto a shared reproducing kernel Hilbert space, the model can better distinguish between the two based on

contextual clues. Extensive experiments demonstrate that MePe achieves state-of-the-art performance on the MChIRC dataset.

CCS Concepts

• Computing methodologies → Lexical semantics.

Keywords

Multimodal Chinese idiom, Reading Comprehension, Metaphor

ACM Reference Format:

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1 Introduction

Chinese idioms are the crystallized linguistic essence of Chinese culture, usually composed of four Chinese characters. For example, “水泄不通” (The water can not even flow) in Table 1 can express both its literal meaning and its metaphorical implication (Be so jammed as to be impassable). Due to the discrepancy between their literal and metaphorical meanings, it poses a tremendous challenge for existing models to understand Chinese idioms, even for existing large language models [11, 42]. The metaphorical information contained in the images of Chinese idioms in multimodal forms further intensifies the challenge of understanding multimodal Chinese idioms. Accurate understanding of Chinese idioms in multimodal forms can significantly improve the understanding of Chinese cultural symbols by artificial intelligence systems, and also promote the research of downstream tasks such as machine translation [27, 30], image generation [1, 20], and sentiment analysis [36, 41].

Existing research on Chinese idiom reading comprehension is mainly categorized into unimodal and multimodal according to their modality types. The most landmark unimodal work is the Chinese idiom cloze task proposed by Zheng et al. [50], who established the widely used Chinese idiom cloze dataset ChID. Next, some research

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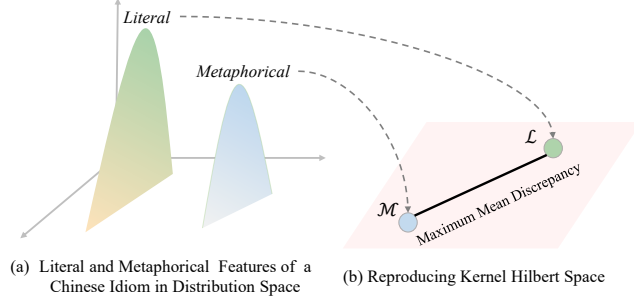
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Table 1: An example of a Chinese idiom where the literal meaning and the metaphorical meaning are discrepant.

Idiom	水泄不通
Literal	The water can not even flow.
Metaphorical	Be so jammed as to be impassable.

**Figure 1: Intuitively visualizing a Chinese idiom’s literal and metaphorical semantic features via Maximum Mean Discrepancy to enhance understanding of its “dual semantics”.**

[9, 24, 33, 37, 43, 45] proposed various methods to achieve good performance on the ChID dataset. However, as shown in Figure 2, the metaphorical mapping of idioms like “The water can not even flow” to scenes like “ducks congested road” is hard to explain through text alone, as its meaning relies on the visual context of congestion. Therefore, Wang et al. [42] proposed the first dataset for the multimodal Chinese idiom reading comprehension task, which fills the gap in this research. They also proposed a dual-contrastive learning network to align contextual semantics and images, achieving a performance that outperforms MLLMs.

Although some advances have been made in the study of multimodal Chinese idiom reading comprehension tasks, there are still two key challenges in this task: **(1) There is a discrepancy between the literal and metaphorical meanings of Chinese idioms.** As shown in Table 1, the idiom “The water can not even flow” is often used to describe a very crowded or tightly fenced area, but the actual meaning of this Chinese idiom has little to do with “water” itself. **(2) The same Chinese idiom has different meanings in different scenarios**, which requires the targeted understanding of experts who specialize in different fields. Those who have not studied Chinese idioms systematically grasp only the literal meaning of this idiom in Table 1. Those with a Chinese background can understand its metaphorical meaning.

To address the above challenges, we propose a framework named MePe to rethink the solution to the multimodal idiom reading comprehension task. Firstly, we propose a literal metaphorical semantic graph that systematically transforms the implicit discrepancy between the literal and metaphorical meanings of Chinese idioms into structured explicit relationships, so that the metaphorical meanings can be understood more easily through the association with the literal meanings. Then, we design a mixture of idiom experts comprising a literal idiom expert and a metaphorical idiom expert that

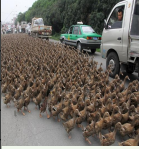
Text	上万只鸭子成群结队过马路，马路瞬间 #idiom#。 Tens of thousands of ducks crossed the road in groups, and the road was instantly #idiom#.	Image
Candidate Idioms	NO.1 摩肩接踵 Shoulder to shoulder and heel to heel	
	NO.2 水泄不通 Be so jammed as to be impassable ✓	
	NO.3 土豪劣绅 Local tyrants and evil gentry	
	NO.4 严阵以待 Be ready in full battle array	
	NO.5 人满为患 Packed with people	
	NO.6 心灰意懒 Loss of confidence and demoralization	
	NO.7 费尽心机 Rack one's brains in scheming	
		正确选项是 NO.2 The correct choice is NO.2

Figure 2: An example of multimodal Chinese idiom reading comprehension task.

specialize in extracting their respective semantic cues. Their complementary outputs are then fused to achieve an understanding of the dual semantics of Chinese idioms in different scenarios. Finally, we employ the maximum mean discrepancy to control the degree of difference between the literal and metaphorical meanings of idioms as shown in Figure 1. By mapping literal and metaphorical features to a shared reproducing kernel Hilbert space, the model can learn discriminative semantic representations to improve the accuracy of idiom understanding. Extensive experiments demonstrate that MePe achieves state-of-the-art performance.

The main contributions are as follows:

- We explore multimodal Chinese idiom reading comprehension tasks from a metaphorical perspective for the first time and propose a novel framework named MePe.
- We propose a literal metaphorical semantic graph in which the literal and metaphorical meanings of each idiom are treated as separate nodes. By defining bi-relational edges and aggregating neighbor information, the model jointly captures the topological interplay of both semantics.
- We propose a mixture of idiom experts consisting of a literal idiom expert and a metaphorical idiom expert. Their division of labor matches idioms’ dual attributes, enabling understanding of Chinese idioms across scenarios.
- We incorporate the maximum mean discrepancy constraint to control divergence between literal and metaphorical feature distributions, enabling the model to understand Chinese idioms accurately across contexts.

2 Related Work

Chinese Idiom Reading Comprehension. The task of Chinese idiom comprehension is intended to enable models to grasp the precise meaning of idioms. Cui et al. [6, 7] employed the full representation of queries and the “Overly Attentive Reader” to tackle reading comprehension tasks. Zheng et al. [50] proposed the ChID dataset for reading Chinese idioms and, based on this dataset, put forward three baseline methods: LM, AR, and SAR. Based on ChID datasets, Long et al. [24] and Wang et al. [43] using an attention mechanism, Tan and Jiang [37] using a two-stage model, and Dai et al. [9] and Sha et al. [33] using interpretive methods, all of which significantly improve the machine’s understanding of Chinese idioms. Wu et al. [45, 46] employed contrastive learning to achieve state-of-the-art performance on the ChID dataset. Wang et al. [42]

proposed the MChIRC dataset, which fills the gap in multimodal Chinese idiom reading comprehension tasks. They also proposed the DCIGN to align multimodal features from coarse to fine. However, they overlooked the inherent traits of Chinese idioms. In contrast to these methods, we make the first attempt to address this task from a metaphorical perspective.

3 The MePe Model

3.1 The Overall Framework

The framework of MePe is present in Figure 3. It consists of four components: Multimodal Feature Extraction, Literal Metaphorical Semantic Graph, Mixture of Idiom Experts, and Prediction Module.

3.2 Multimodal Feature Extraction

The input passage for the model is denoted as $P = \{w_1, \dots, [MASK], \dots, w_n\}$, where each character w_n represents a Chinese character, and the position of the idiom is marked with $[MASK]$. The input image for the model is a corresponding image of the idiom, denoted as $I = \{image\}$. We employ a pre-trained BERT¹ to extract text features and a pre-trained DeiT² to extract image features.

$$\begin{cases} T_o = BERT([CLS], P, [SEP]), \\ V_o = DeiT(I), \end{cases} \quad (1)$$

where T_o represents extracted primitive text features, V_o represents extracted primitive image features, and the special marks $[CLS]$ and $[SEP]$ are boundary marks for guiding and terminating input.

To achieve multimodal contextual semantic consistency modeling, we employ the dual-contrastive learning loss from coarse to fine [42], which aligns and fuses multimodal features to bring text-image consistency semantics closer and to exclude interferences. Thereby, modeling the true meaning of idioms in specific contexts.

$$\begin{cases} \tilde{T}_o = LN(SA(Q_{T_o}, K_{T_o}, V_{T_o}) + T_o), \\ \tilde{V}_o = LN(SA(Q_{V_o}, K_{V_o}, V_{V_o}) + V_o), \end{cases} \quad (2)$$

where $\tilde{T}_o$ and $\tilde{V}_o$ are the aligned text features and image features, respectively. LN denotes layer normalization operation, and SA denotes self-attention operation. Then, we fuse the extracted refined features with the extracted original features to obtain the multimodal features $\mathcal{F}_M$ as follows:

$$\mathcal{F}_M = T_o + V_o + \tilde{T}_o + \tilde{V}_o, \quad \mathcal{F}_M \in \mathbb{R}^{B \times D}, \quad (3)$$

where B denotes batch size, and D denotes feature dimension.

We compute the coarse and fine contrastive learning loss.

$$\begin{cases} \mathcal{L}_{ccl} = -\sum_{i=1}^N \log \frac{e^{\text{sim}(T_o^i, V_o^i)/\tau}}{\sum_{j=1}^N e^{\text{sim}(T_o^i, V_o^j)/\tau}}, \\ \mathcal{L}_{fcl} = -\sum_{i=1}^N \log \frac{e^{\text{sim}(\tilde{T}_o^i, \tilde{V}_o^i)/\tau}}{\sum_{j=1}^N e^{\text{sim}(\tilde{T}_o^i, \tilde{V}_o^j)/\tau}}, \end{cases} \quad (4)$$

where N represents the batch size, T_o^i and V_o^i denote the i_{th} sample comes from two different modal features. sim is the cosine similarity. τ is the temperature hyperparameter. We obtain the final dual contrastive learning loss $\mathcal{L}_{dcl}$.

$$\mathcal{L}_{dcl} = u\mathcal{L}_{ccl} + v\mathcal{L}_{fcl}, \quad (5)$$

where u and v are the weights of the coarse contrastive learning loss and the fine contrastive learning loss, respectively. We set u to 0.6 and v to 0.4, consistent with the settings of DCIGN [42].

3.3 Literal Metaphorical Semantic Graph

To distinguish between the literal and metaphorical meanings of Chinese idioms, we propose a literal metaphorical semantic graph that converts the implicit discrepancy between literal and metaphorical meanings into structured explicit relationships, thereby making the metaphorical meanings easily understandable through their association with the literal ones.

Node Relationship Form for Literal Metaphorical Semantic Graph. To show the relationship between the nodes in the literal metaphorical semantic graph more intuitively, we present a table of node relationships as shown in Figure 4. The node relationship form for the literal metaphorical semantic graph constructs a fourteen-node topology comprising seven literal nodes (L_1-L_7) and seven metaphorical nodes (M_1-M_7), with three edge types: self-loops (preserving intrinsic node features), literal Metaphorical (L-M) connections (modeling intra-idiom dual-meaning interactions), and full metaphorical connections (enabling cross-idiom metaphorical knowledge transfer). This structure embeds linguistic priors that capture the dual-meaning nature of Chinese idioms and the abstract commonality of metaphors, while retaining adaptability through learnable edge weights. For multimodal idiom comprehension tasks, it enhances feature representation by fusing literal details and metaphorical abstractions within individual idioms, leverages the inter-metaphorical network for cross-idiom inference, and uses L-M edges as multimodal alignment bridges. Self-loops ensure semantic stability, and the integration of prior knowledge balances inductive bias and data-driven adaptation. This design addresses the core semantic challenges of comprehending Chinese idioms, thereby improving performance on a multimodal Chinese idiom reading comprehension task.

Specifically, we first utilize a pre-trained BERT model to extract features for each Chinese idiom within the feature candidate set.

$$C_k = BERT(Idiom_k), \quad C = \text{Concat}(C_1, \dots, C_k), \quad (6)$$

where k stands for the k_{th} idiom, $k \in [1, 7]$, $C \in \mathbb{R}^{B \times N \times D}$, B as the batch size, N as the number of idioms in each example, D as the hidden dimension ($D = 768$). We use linear projection to map the features of candidate idioms to the literal and the metaphorical semantic space.

$$\begin{cases} L = C_k W_{lit} + b_{lit}, & L \in \mathbb{R}^{B \times N \times D}, \\ M = C_k W_{met} + b_{met}, & M \in \mathbb{R}^{B \times N \times D}, \end{cases} \quad (7)$$

where W_{lit} and $W_{met} \in \mathbb{R}^{D \times D}$ are learnable weight matrices, b_{lit} and $b_{met} \in \mathbb{R}^D$ are bias vectors. We then construct a literal metaphorical semantic graph by following five stages.

Stage I: Node Selection. We concatenate literal and metaphorical features into a unified graph structure and select literal and metaphorical nodes for each candidate idiom.

$$\mathcal{H} = \text{Concat}(L; M), \quad \mathcal{H} \in \mathbb{R}^{B \times (2N) \times D}. \quad (8)$$

The metaphorical semantic nodes M_n and literal semantic nodes L_n are selected, $n = [1, 7]$.

¹<https://huggingface.co/hfl/chinese-bert-wwm>

²<https://huggingface.co/facebook/deit-base-patch16-224>

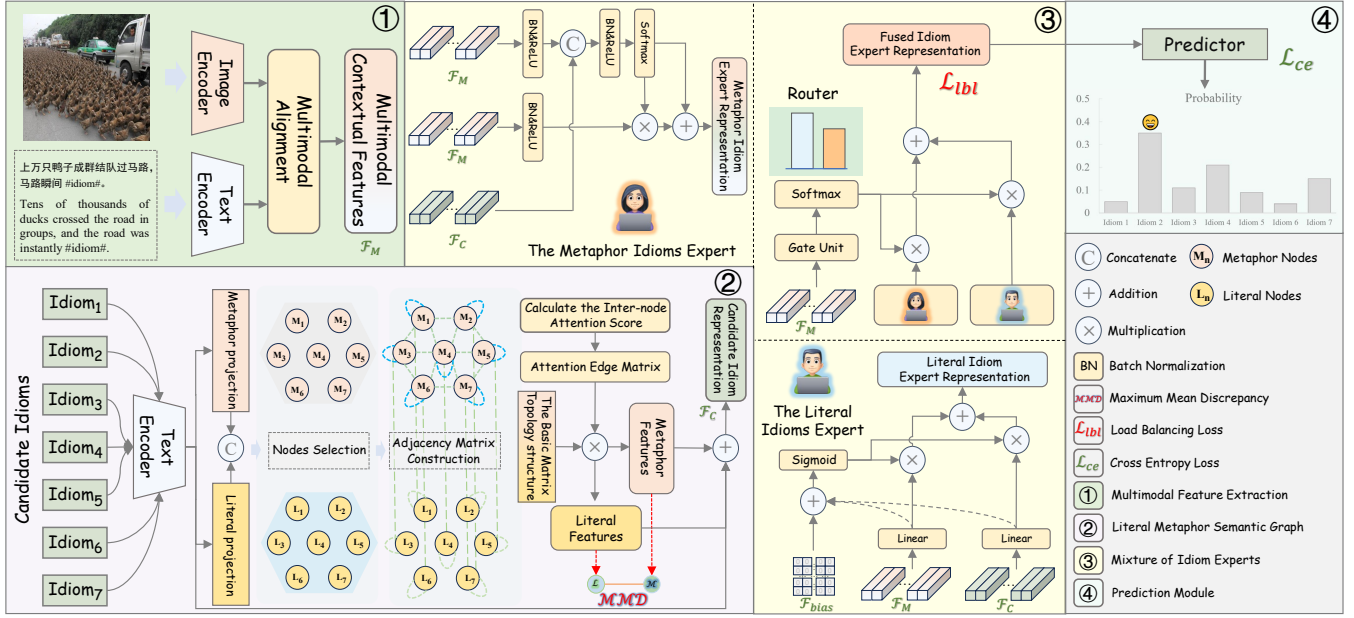


Figure 3: The framework of MePe.

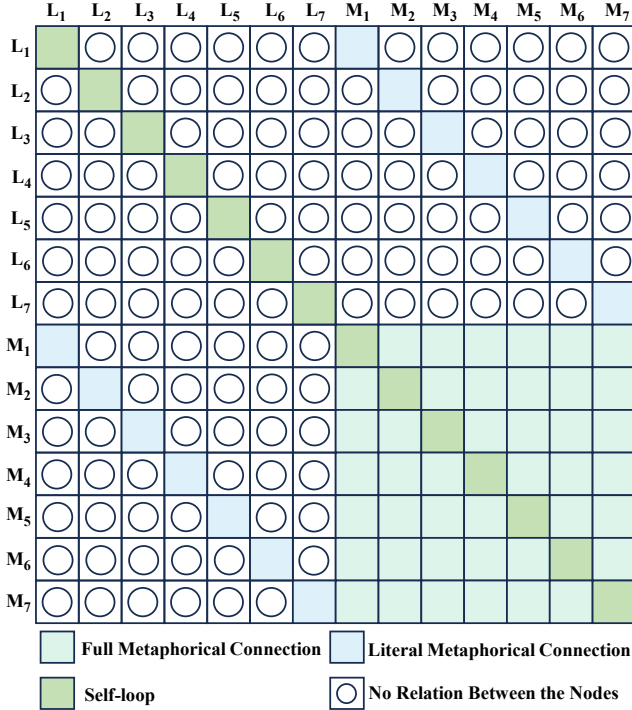


Figure 4: Node relationship form for literal metaphorical semantic graph.

Stage II: Adjacency Matrix Construction. The detailed correspondence between nodes in the literal metaphorical semantic adjacency matrix is shown in Figure 4. The initial adjacency matrix

$\mathcal{A}^{(0)} \in \{0, 1\}^{2N \times 2N}$ is defined by

$$\mathcal{A}_{ij}^{(0)} = \begin{cases} 1, & i = j, \\ 1, & i \in [1, N], j = i + N, \\ 1, & i \in [N + 1, 2N], j \in [N + 1, 2N], i \neq j, \\ 0, & \text{otherwise,} \end{cases} \quad (9)$$

where 1 indicates that there is a node relationship, and 0 indicates that there is no node relationship.

Stage III: Attention Score Computation. In order to allocate separate representation spaces for the relationships between literal and metaphorical nodes, we perform an exclusive linear mapping of the relationships between each type of node and reshape them into a multi-head parallel structure. For each relation $r = (1, \dots, R)$, we apply a linear layer.

$$Z^{(r)} = HW_r + b_r, \quad Z^{(r)} \in \mathbb{R}^{B \times (2N) \times (HD)}, \quad (10)$$

where $H=8$ denotes the number of attention heads, $D=d/H$ denotes the per-head dimension. We then reshape and transpose to heads: $Z^{(r)} \rightarrow \tilde{Z}^{(r)}, \tilde{Z}^{(r)} \in \mathbb{R}^{B \times H \times (2N) \times D}$. For each head h and pair of nodes (i, j) within relation r ,

$$e_{t,h,i,j}^{(r)} = \text{LeakyReLU} \left(a_r^\top \text{concat} [\tilde{Z}_{t,h,i,:}^{(r)}; \tilde{Z}_{t,h,j,:}^{(r)}] \right), \quad (11)$$

where concat denotes concatenation operation and $a_r \in \mathbb{R}^{2D}$ is the learned attention vector for relation r . $e_{t,h,i,j}^{(r)}$ denotes the unnormalized attention score for relation r , example t in the batch, head h , from node i to node j .

Stage IV: Attention Edge Matrix. We also learn a scalar weight for each potential edge (i, j) .

$$\varepsilon_{t,i,j} = \sigma \left(\text{MLP}(\text{concat} [H_{t,i,:}; H_{t,j,:}]) \right), \quad (12)$$

where MLP is a two-layer perceptron with a sigmoid function σ . These weights form an attention edge matrix $\varepsilon_{t,i,j}^{(r)} \in \mathbb{R}^{B \times N \times N}$, and $:$ denotes all elements of the dimension.

Stage V: Updating Literal Metaphorical Semantic Features. We combine the initial adjacency $\mathcal{A}^{(0)}$ with the attention edge matrix $\varepsilon_{t,i,j}^{(r)}$ to get the weighted adjacency matrix for each Chinese idiom under each relation.

$$\tilde{\mathcal{A}}_{t,i,j}^{(r)} = \mathcal{A}_{t,i,j}^{(0)} \cdot \varepsilon_{t,i,j}^{(r)}. \quad (13)$$

Then we combine $\tilde{\mathcal{A}}_{t,i,j}^{(r)}$ to mask multi-head attention scores, use a softmax function, and aggregate messages from each head and node to update node features under each relationship.

$$\alpha_{t,h,i,j}^{(r)} = \text{Softmax}(e_{t,h,i,j}^{(r)} \tilde{\mathcal{A}}_{t,i,j}^{(r)}). \quad (14)$$

$$\mathbf{m}_{t,i}^{(r)} = \sum_{h=1}^H \sum_{j=1}^{2N} \alpha_{t,h,i,j}^{(r)} \tilde{\mathcal{Z}}_{t,h,j}^{(r)}. \quad (15)$$

Stacking over B and $2N$ yields $\mathbf{M} \in \mathbb{R}^{B \times (2N) \times D}$. After completing message passing on the literal metaphorical semantic graph, the literal semantic features $\mathbf{U}_{\text{lit}}$ and metaphorical semantic features $\mathbf{U}_{\text{met}}$ of the corresponding nodes are extracted via indexed slices $\mathbf{M}$.

$$\begin{cases} \mathbf{U}_{\text{lit}} = \mathbf{M}_{:, 1 \dots N, :}, & \mathbf{U}_{\text{lit}} \in \mathbb{R}^{B \times N \times D}, \\ \mathbf{U}_{\text{met}} = \mathbf{M}_{:, N+1 \dots 2N, :}, & \mathbf{U}_{\text{met}} \in \mathbb{R}^{B \times N \times D}. \end{cases} \quad (16)$$

Finally, we connect literal features $\mathbf{U}_{\text{lit}}$ and the metaphorical features $\mathbf{U}_{\text{met}}$ with the original candidate idiom features $\mathbf{C}$ by residuals, and then by layer normalization LN , we get the candidate idiom features $\mathcal{F}_C$ after literal metaphorical semantic graph enhancement as follows.

$$\mathcal{F}_C = LN(\mathbf{U}_{\text{lit}} + \mathbf{U}_{\text{met}} + \mathbf{C}), \quad \mathcal{F}_C \in \mathbb{R}^{B \times N \times D}. \quad (17)$$

3.4 Maximum Mean Discrepancy

The literal and metaphorical meanings of a Chinese idiom are essentially two semantic-level expressions of the same idiom, as shown in Figure 1. They must share a common semantic foundation while remaining distinct in style or higher-level semantics. Therefore, we apply the maximum mean discrepancy [16] constraint to simultaneously extract literal and metaphorical features, transforming them to the reproducing Hilbert space to ensure they retain a shared base while remaining separable at the stylistic or high-level semantic layer. We first flatten the batch and sequence dimensions to obtain the sample matrix.

$$\begin{cases} \mathbf{U}_{\text{lit}} \in \mathbb{R}^{B \times N \times D} \xrightarrow{\text{flatten}} \tilde{\mathbf{U}}_{\text{lit}} \in \mathbb{R}^{(B \cdot N) \times D}, \\ \mathbf{U}_{\text{met}} \in \mathbb{R}^{B \times N \times D} \xrightarrow{\text{flatten}} \tilde{\mathbf{U}}_{\text{met}} \in \mathbb{R}^{(B \cdot N) \times D}. \end{cases} \quad (18)$$

We use a Gaussian kernel $K(\mathbf{u}, \mathbf{v}) = \exp(-\gamma \|\mathbf{u} - \mathbf{v}\|_2^2)$ to measure the similarity between any two literal or metaphorical features of Chinese idioms. We set the bandwidth $b \in [0.5, 2.0]$ and the standard Gaussian kernel function $\gamma = \frac{1}{2b^2}$. The kernel matrix is constructed as follows.

$$\begin{cases} [K_{\text{lit}, \text{lit}}]_{i,j} = K(\tilde{\mathbf{U}}_{\text{lit}_i}, \tilde{\mathbf{U}}_{\text{lit}_j}), K_{\text{lit}, \text{lit}} \in \mathbb{R}^{N \times N}, \\ [K_{\text{met}, \text{met}}]_{i,j} = K(\tilde{\mathbf{U}}_{\text{met}_i}, \tilde{\mathbf{U}}_{\text{met}_j}), K_{\text{met}, \text{met}} \in \mathbb{R}^{N \times N}, \\ [K_{\text{lit}, \text{met}}]_{i,j} = K(\tilde{\mathbf{U}}_{\text{lit}_i}, \tilde{\mathbf{U}}_{\text{met}_j}), K_{\text{lit}, \text{met}} \in \mathbb{R}^{N \times N}. \end{cases} \quad (19)$$

Let $\mathbf{I}_N$ be the identity matrix $N \times N$ (self-comparisons). Define masks: $M_{\text{off}} = \mathbf{I}_N$, $M_{\text{diag}} = \mathbf{I}_N$. Using the above kernel matrices, the maximum mean discrepancy distance with tolerance is calculated by

$$\begin{aligned} \mathcal{L}_{\text{mmd}} &= \text{MMD}_{\eta}^2(\tilde{\mathbf{U}}_{\text{lit}}, \tilde{\mathbf{U}}_{\text{met}}) \\ &= \frac{\sum (K_{\tilde{\mathbf{U}}_{\text{lit}} \tilde{\mathbf{U}}_{\text{lit}}} \odot M_{\text{off}}) + \eta \sum (K_{\tilde{\mathbf{U}}_{\text{lit}} \tilde{\mathbf{U}}_{\text{lit}}} \odot M_{\text{diag}})}{N(N-1+\eta)} \\ &\quad + \frac{\sum (K_{\tilde{\mathbf{U}}_{\text{met}} \tilde{\mathbf{U}}_{\text{met}}} \odot M_{\text{off}}) + \eta \sum (K_{\tilde{\mathbf{U}}_{\text{met}} \tilde{\mathbf{U}}_{\text{met}}} \odot M_{\text{diag}})}{N(N-1+\eta)} \\ &\quad - 2 \frac{\sum K_{\tilde{\mathbf{U}}_{\text{lit}} \tilde{\mathbf{U}}_{\text{met}}}}{N^2}, \end{aligned} \quad (20)$$

where $\eta \in [0.1, 0.5]$ is a tolerance parameter and $\odot$ denotes element-wise multiplication.

3.5 Mixture of Idiom Experts

The same Chinese idiom has different meanings in different scenarios, which requires a targeted understanding by experts who specialize in different fields. Based on this finding, we propose a mixture of idiom experts. The literal idiom expert uses a sigmoid function to perform a weighted fusion of candidate idiom features and contextual features, thereby capturing literal semantic features. The metaphorical idiom expert uses a more sophisticated attention mechanism to capture metaphorical associations between idioms and contexts through the concatenation of candidate idiom and contextual features, along with attentional weighting [28].

Literal Idiom Expert. We define the learnable parameters $(W_g, U_g) \in \mathbb{R}^{D \times D}$ and $\mathcal{F}_{\text{bias}} \in \mathbb{R}^D$. The gating coefficients are calculated by a sigmoid function σ .

$$G = \sigma(W_g \mathcal{F}_C + U_g \mathcal{F}_M + \mathcal{F}_{\text{bias}}), \quad G \in \mathbb{R}^{B \times N \times D}. \quad (21)$$

We get the representations of the literal expert as follows.

$$L_o = G \cdot \mathcal{F}_C + (1 - G) \cdot \mathcal{F}_M, \quad L_o \in \mathbb{R}^{B \times N \times D}. \quad (22)$$

Metaphorical Idiom Expert. We define three learning mappings.

$$\begin{cases} \phi_a(\cdot) = \text{ReLU}(BN(\cdot)), \\ \phi_k(\mathcal{F}_M) = \phi_a(\mathcal{F}_M), \\ \phi_k(\mathcal{F}_M) = \phi_a(\mathcal{F}_M). \end{cases} \quad (23)$$

The metaphorical representations M_o of Chinese idioms are obtained sequentially through the following operations.

$$\begin{cases} \tilde{\mathcal{F}}_M = \text{Concat}((\phi_k(\mathcal{F}_M)); \mathcal{F}_C), \\ \tilde{\phi}_k = \text{Softmax}(\phi_a(\tilde{\mathcal{F}}_M)), \\ M_o = \tilde{\phi}_k \cdot \phi_o(\mathcal{F}_M) + \tilde{\phi}_k. \end{cases} \quad (24)$$

Expert Features Fusion. We fuse literal idiom features with metaphorical idiom features by a router. We first compute the gating weights $\mathbf{w}^{(b)} = \text{Softmax}(f_{\text{gate}}(\mathcal{F}_M^{(b)}))$. f_{gate} is a gate unit composed of two linear layers. Then, we get the final feature representation $\mathcal{F}_{\text{final}}$.

$$\mathcal{F}_{\text{final}} = \mathbf{w}^{(b)} M_o + \mathbf{w}^{(b)} L_o. \quad (25)$$

The Kullback-Leibler divergence is used as a loss function to measure the ideal uniform expert use distribution $\mathbf{p}$ and the actual

Table 2: Comparison results (%) with baseline models on the MChIRC dataset. “†” stands for rerunning these methods.

Modality	Method	Dev-Acc	Test-Acc	Sim-Acc	Out-Acc	Avg-Acc
Text	BERT-WWM [5]	63.43	60.40	59.50	49.79	58.28
	RoBERTa [5]	64.49	61.85	60.50	50.72	59.39
	macBERT [5]	64.27	61.43	60.20	50.31	59.05
	PRIEM† [33]	65.94	64.24	71.30	47.13	62.15
	RISCF† [46]	67.02	66.12	66.65	46.82	61.65
	MSCLM† [45]	71.19	69.56	67.89	51.95	65.15
Image	VGG-16 [35]	45.70	44.05	45.27	33.36	42.10
	CLIP [32]	40.66	38.63	42.57	26.35	37.05
	DeiT [40]	55.46	54.01	54.81	40.64	51.23
Multimodal	MSCA† [18]	68.63	69.15	68.01	55.04	65.21
	Multi-view CLIP† [31]	59.05	61.05	51.04	49.24	55.10
	DivE† [22]	74.24	71.56	66.71	56.03	67.13
	CLTL† [44]	65.55	64.05	60.24	50.75	60.15
	DCIGN† [42]	77.26	77.16	77.34	60.25	73.00
	MePe (Ours)	83.69_(+6.43)	83.09_(+5.93)	81.63_(+4.29)	72.18_(+11.93)	80.15_(+7.15)

average gating weights $\bar{w}$.

$$\mathcal{L}_{\text{lbl}} = \text{KL}(\bar{w} \parallel p) = \sum_{i=1}^2 \bar{w}_i \log\left(\frac{\bar{w}_i}{p_i}\right). \quad (26)$$

3.6 Prediction Module

We input the final features $\mathcal{F}_{\text{final}}$ after the mixture of idiom experts into the linear layer, then use the function to probabilistically predict the seven candidate idioms in the candidate set.

$$P(\text{Idiom}_k | P, I) = \text{Softmax}(W\mathcal{F}_{\text{final}} + b), \quad (27)$$

where Idiom_k stands for the k_{th} idiom in the candidate set. We calculate the difference between the predicted probability distribution of the correct idiom and the true distribution by minimizing the cross-entropy loss function as follows:

$$\mathcal{L}_{ce} = - \sum_{k=1}^7 C_g \log(P(\text{Idiom}_k | P, I)), \quad (28)$$

where C_g stands for the one-hot label distribution for the correct idiom. The final total loss we get is shown below.

$$\mathcal{L} = \mathcal{L}_{ce} + \mathcal{L}_{dcl} + \alpha \mathcal{L}_{mmd} + \beta \mathcal{L}_{lbl}, \quad (29)$$

where α and β are the weights of the $\mathcal{L}_{mmd}$ loss and the $\mathcal{L}_{lbl}$ loss, respectively.

4 Datasets and Experimental Setup

Datasets. All experiments are conducted on the multimodal Chinese idiom dataset (MChIRC), which is currently the only multimodal Chinese idiom dataset. A detailed introduction to the MChIRC dataset is provided in the Appendix B.

Evaluation Metrics. We adopt accuracy as the evaluation metric, which is defined as the proportion of test samples where the idiom selected by the model matches the correct one. Additionally, Avg-Acc denotes the average accuracy across all previous test sets and is used to assess the overall performance of the model.

Baselines. We compare six text modality methods, three image modality methods, four classical multimodal methods, and one

method specifically designed to solve the multimodal Chinese idiom reading comprehension task. More detailed descriptions of these compared methods are provided in the Appendix D.

Implementation Details. For the text modality. We set the maximum sequence length to 128, and the hidden feature dimension is set to 768. For the image modality, we scale the image size to 224×224 and set the hidden layer to 768. We use a batch size of 32 and the Adam optimizer. All experiments are run on an NVIDIA Tesla A100 GPU with 40 GB of memory. We use Transformers 4.29.2 and Pytorch 1.7.1. We set the total training epochs of the model to 20. We set the random seed to 42 for reproducibility. The learning rate is set to 5e-5. We use an early-stop mechanism [39] with an early-stop count of 2.

5 Results and Analysis

5.1 Comparison with Baselines

We compare the performance of our proposed MePe with the comparison methods, and the results are shown in Table 2. Among unimodal methods, text-modal approaches consistently outperform image-modal ones. We analyze this because the core of Chinese idioms lies in abstract semantics, which can be directly conveyed by words. In contrast, images are limited by their concrete expressive nature, making it difficult to effectively map the deeper semantics of idioms. Multimodal methods effectively make up for the limitations of a single modality by aligning and fusing the modal information of text and images in the latent space, thereby capturing the deep semantics of Chinese idioms more comprehensively. Our proposed MePe demonstrates significant advantages over other multimodal methods. Compared to classical multimodal methods, our MePe achieves much higher accuracy across all Dev, Test, Sim, and Out sets. Compared to DCIGN, the current state-of-the-art method, MePe achieves accuracy improvements of 6.43%, 5.93%, 4.29%, and 11.93% on the Dev, Test, Sim, and Out sets, respectively, with an average accuracy improvement of 7.15%. We attribute this advantage to the following reasons: Although DCIGN employs dual contrastive learning loss from coarse to fine to adjust image and

Table 3: Comparison results of the MLLMs with 500 samples in the Out set, “B” stands for billion.

Method	Parameters	Acc	Team
GLM4V	9B	52.8	ZhipuAI
CogVLM2	8B	49.8	ZhipuAI
InternVL2-Pro	-	51.8	OpenGVLab
Qwen1.5	110B	52.6	Aliyun
GPT-4o	175B	53.8	OpenAI
GPT-5	-	62.8	OpenAI
MePe (Ours)	-	69.2	-

text features, it ignores the essential attribute that idioms have both literal and metaphorical meanings, which is crucial for a comprehensive understanding of the connotations of idioms. In contrast, MePe constructs a literal metaphorical semantic graph, enabling it to fully capture the potential connections between the literal and metaphorical semantics of idioms. Furthermore, MePe establishes an expert team consisting of a metaphorical idiom expert and a literal idiom expert, who are specifically responsible for extracting their respective semantic clues. By fusing these clues, MePe achieves precise matching of the dual attributes of Chinese idioms. Finally, MePe employs the maximum mean discrepancy to constrain the literal and metaphorical features of Chinese idioms to further control the difference between these two feature distributions and ensure balanced dual-semantic learning.

5.2 Comparison with MLLMs

To demonstrate the superiority of our method, we compare it with the six multimodal large language models. The introduction of these MLLMs is in Appendix H, and the prompt template is in Appendix I. We evaluate MePe on the Out set, which is designed to assess idiom comprehension under real-world multimodal scenarios. As shown in Table 3, general-purpose MLLMs such as GLM4V, InternVL2-Pro, Qwen1.5, and GPT-4o achieve only moderate performance (49.8%–53.8%), while even GPT-5 reaches just 62.8% accuracy. In stark contrast, our proposed MePe achieves 69.2%, outperforming all baselines by a substantial 6.4% absolute margin. This performance gain demonstrates that large-scale parameterization alone is insufficient for modeling the dual semantics of Chinese idioms. MePe’s performance gains derive from three core design innovations: We propose a literal metaphorical semantic graph that structurally captures the interplay between an idiom’s literal and metaphorical meanings. We propose a mixture of idiom experts that enables semantic specialization through targeted collaboration between experts focused on literal and metaphorical inference. The maximum mean discrepancy constraint bolsters robustness against context-driven semantic shifts. These three components form an organic synergy that, from modeling semantic structures to enabling specialized division of labor and regulating cross-context stability, jointly address the core challenges of associating, distinguishing, and adapting literal and metaphorical meanings in multimodal Chinese idiom comprehension.

Table 4: The performance of different components on the MChIRC dataset. ‘w/o’ is an abbreviation for ‘without’.

Method	Dev	Test	Sim	Out	Avg
w/o LMSG	77.48	76.45	78.52	64.11	74.14
w/o LN	79.25	79.05	78.69	66.17	75.79
w/o MN	79.04	78.20	79.84	64.42	75.37
w/o MoIE	77.90	76.39	78.69	63.36	76.21
w/o LE	80.44	78.78	80.11	65.52	74.09
w/o ME	80.15	79.55	78.82	65.31	75.96
w/o $\mathcal{L}_{lbl}$	81.16	79.37	80.46	66.13	76.78
w/o $\mathcal{L}_{mmd}$	79.98	78.38	79.67	65.21	75.81
MePe (Ours)	83.69	83.09	81.63	72.18	80.15

5.3 Ablation Study

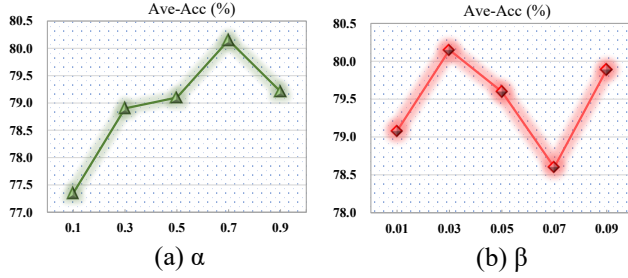
We conduct a series of ablation experiments on various components of the MePe to verify the effectiveness of each component, and the results are shown in Table 4. **(1) When we remove the literal metaphorical semantic graph (LMSG)**, the average accuracy drops by 6.01%. Meanwhile, the average accuracy of the model decreases by 4.36% and 4.78% when we remove the literal nodes (LN) and metaphorical nodes (MN), respectively. This indicates that LMSG can mine the literal and metaphorical meanings in Chinese idioms by encoding the literal and metaphorical knowledge of idioms into learnable graph nodes, and then interacting with the information between the nodes. **(2) When we remove the mixture of idiom experts (MoIE)**, the average accuracy drops by 3.94%. Meanwhile, when the literal idiom expert (LM) and metaphorical idiom expert (ME) are removed, the average accuracy of the model decreases by 6.06% and 4.19%, respectively. This indicates that dividing the literal idiom expert and metaphorical idiom expert to learn the literal and metaphorical semantics of Chinese idioms, respectively, avoids the limitation of a single perspective, thus providing a more comprehensive understanding of Chinese idioms. **(3) When we remove the load balancing loss $\mathcal{L}_{lbl}$** , the average accuracy drops by 3.37%. This indicates that ensuring that both literal idiom expert and metaphorical idiom expert are adequately trained through the use of $\mathcal{L}_{lbl}$ explicit constraints maintains the diversity of dynamic routing at the idiomatic level. **(4) When we remove the maximum mean discrepancy $\mathcal{L}_{mmd}$** , the average accuracy drops by 4.34%. This indicates that both literal and metaphorical features share a consistent semantic space under the constraints of MMD while retaining their respective characteristics.

5.4 Hyperparameter Analysis

We perform a hyperparametric analysis of the tolerance η in the maximum mean discrepancy and the bandwidth b of the Gaussian kernel function. The results are shown in Table 5. The experimental results show that when $\eta=0.3$ and $b=1.0$, the model achieves 83.09% accuracy on the test set, 81.63% on the Sim set, and 72.18% on the Out set, with an average accuracy of 80.15%, which is the best performance. This indicates that a moderate η can balance the literal and metaphorical semantics of Chinese idioms, while too small an η will make the alignment rigid, and too large an η will affect generalization. Similarly, a kernel bandwidth that is too narrow amplifies noise, while a kernel that is too wide blurs the

Table 5: The performance of different values of η and b .

η, b	Dev	Test	Sim	Out	Avg
$\eta=0.1$	83.50	82.74	81.22	70.95	79.60
$\eta=0.2$	83.84	82.45	80.47	70.61	79.34
$\eta=0.3$	83.69	83.09	81.63	72.18	80.15
$\eta=0.4$	83.35	81.23	80.39	70.71	78.92
$\eta=0.5$	84.44	82.50	81.23	71.26	79.86
$b=0.5$	81.58	81.06	80.80	70.75	78.55
$b=1.0$	83.69	83.09	81.63	72.18	80.15
$b=1.5$	83.02	81.66	80.95	71.57	79.30
$b=2.0$	82.15	81.99	80.02	71.57	78.93

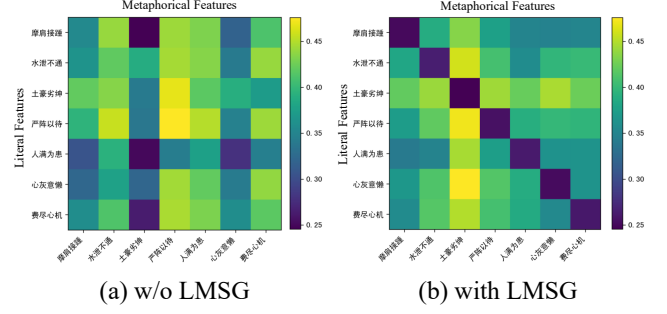
**Figure 5: The performance of different values of α and β .**

semantic difference between the literal and metaphorical semantics of Chinese idioms.

We also conduct a hyperparametric analysis to investigate how the weights α (related to $\mathcal{L}_{mmd}$) and β (related to $\mathcal{L}_{lbi}$) in the total loss function impact the model’s performance. The results are shown in Figure 5. We find that the model achieves its highest average accuracy when α is set to 0.7 and β is set to 0.03. This optimal combination of weights reveals important insights into the role of each loss component: a moderately weighted $\mathcal{L}_{mmd}$ effectively regulates the alignment between literal and metaphorical semantic feature distributions without excessively narrowing the meaningful distinctions that exist between these two semantic layers of idioms, thus preserving the richness of their dual meanings. Similarly, a moderately weighted $\mathcal{L}_{lbi}$ plays a critical role in managing the activation patterns of the various experts within the Mixture of Idiom Experts (MoIE) mechanism. By preventing over-reliance on a subset of experts and encouraging balanced participation across the entire expert team, this loss ensures that the diverse semantic clues each expert specializes in are fully leveraged, ultimately improving the performance of the multimodal Chinese idiom reading comprehension task.

5.5 Visualization

We compare the decoupling heatmaps of the literal and metaphorical semantic features of candidate idioms with and without the literal metaphorical semantic graph (LMSG), as shown in Figure 6. In Figure 6(a), when LMSG is not incorporated, the literal and metaphorical features of the idioms appear disorganized and mutually irrelevant. This chaotic distribution significantly hinders the model’s ability to accurately identify and extract the metaphorical

**Figure 6: The decoupling heatmap of literal and metaphorical features of seven Chinese candidate idioms before and after the literal metaphorical semantic graph. The color is darker, the farther the distance. ‘w/o’ is an abbreviation for ‘without’.**

meanings of the idioms. In contrast, as illustrated in Figure 6(b), by constructing the literal metaphorical semantic graph, we establish explicit edge connections between the literal and metaphorical nodes of idioms within the graph. This effectively constrains the distance between the two types of semantics in the feature space and markedly reduces the interference between literal and metaphorical information. The graph structure facilitates more effective semantic disambiguation, allowing the model to capture nuanced interactions between literal and figurative meanings. As a result, LMSG not only enhances the model’s sensitivity to metaphorical associations conveyed by idioms in specific contexts but also enables a finer-grained distinction between their literal and metaphorical meanings, ultimately improving the performance of the multimodal Chinese idiom reading comprehension task.

6 Conclusion

In this paper, we rethink the solution to the multimodal idiom reading comprehension task from a metaphorical perspective and propose a framework named MePe. Firstly, we construct the literal metaphorical semantic graph, which takes the literal and metaphorical meanings of idioms as dual nodes and learns their topological features by graph structure. Secondly, we design a mixture of idiom experts consisting of a literal idiom expert and a metaphorical idiom expert to focus on the two kinds of semantic cues separately. Finally, we introduce the maximum mean discrepancy constraint to ensure that the metaphorical semantics are not overwhelmed by the literal semantics. Extensive comparative and ablation experiments demonstrate that MePe achieves state-of-the-art performance. We believe that MePe can provide a new perspective and approach to the understanding of other special linguistic phenomena in Chinese in the form of two-part allegorical sayings or poetry.

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Appendix

A Related Work

A.1 Metaphor Understanding

Since Johnson and Lakoff [21] proposed conceptual metaphor theory (CMT), the Metaphors Understanding task pushes the capabilities of deep learning at higher levels of language understanding and generation. Inspired by this theory, Ge et al. [13] proposed the first interpretable metaphor recognition model. Mao et al. [26] proposed a large labeled dataset and MetaPro2.0-P, which strongly contributed to the research on unimodal metaphor understanding. Zhang et al. [49] proposed a multimodal metaphor dataset, MultiMET, which opens the door to automatic metaphor understanding by exploring multimodal cues and their interactions. Chakrabarty et al. [3], He et al. [17], Qian et al. [29] used multimodal large language models to improve the model’s performance on metaphor comprehension. Despite the progress made in the study of metaphors in the general domain, there is still some room for progress in the study of metaphorical understanding of Chinese idioms, a special linguistic phenomenon in Chinese. Therefore, we explore the metaphorical meanings in Chinese idioms from a metaphorical perspective.

A.2 Mixture of Experts

The Mixture of Experts (MoE) model is based on the principle of “divide and conquer” [19], and its core idea consists of an expert network and a gated network with dynamically assigned weights. MoE has spawned innovations in multiple domains: LSTM-based MoE dynamically fused components of dialog systems [23], sparse gating MoE balanced the efficiency and performance of visual-linguistic models [34], and mutual distillation MoE mitigated the problem “narrow field of view” of recommender systems [47]. In the field of large-scale language modeling, Deepseek significantly improves inference efficiency through a sparse architecture of experts [8]. Other research has focused on expert pruning [4, 10, 25, 48]. Inspired by this advantage of MoE, we propose a mixture of idiom

Table 6: The division of the MChIRC dataset. Aside from the different composition of the candidate idiom sets, the data in the Test set and the Sim set are identical.

	Train	Dev	Test/Sim	Out	Total
Idiom	2,926	2,456	2,881	2,926	2,926
Pair	29,054	4,151	8,302	2,926	44,433

experts, which consists of a literal idioms expert and a metaphorical idioms expert. The two experts apply their respective expertise in semantic learning and adaptive fusion through gated networks to comprehensively understand Chinese idioms.

B Dataset

The detailed information of the MChIRC dataset is shown in Table 6. To ensure the balance between inter-class and intra-class data, we divide the dataset into training, validation and testing sets by stratified sampling [2] in the ratio of 7:1:2. The candidate idioms in all three sets contain the golden option, as well as the three Chinese idioms that are semantically most similar to the golden option and the three Chinese idioms that are least similar. In addition, MChIRC also contains more challenging test subsets Out set and Sim set, where Out set consists of non-repeated samples of Chinese idioms, and the candidate idioms of Sim set contain the golden option and the six idioms that are semantically most similar to it.

C Task Definition

The objective of the multimodal Chinese idiom reading comprehension task is to pick the most fitting idiom from a candidate set of seven items (six distractive idioms and one correct idiom), leveraging a given passage segment and its paired image. Take Figure 2 as an example, the passage reads “上万只鸭子成群结队过马路，马路瞬间#idiom# (Tens of thousands of ducks crossed the road in groups, and the road was instantly#idiom#)”. This picture depicts a flock of ducks walking on a road, taking up most of the roadway, with vehicles (e.g., a green car, a white truck, etc.) parked and waiting around them. Therefore, in this example, the correct option for “#idiom#” can be identified as “NO.2 水泄不通 (NO.2 Be so jammed as to be impassable)”, which perfectly captures the essence of the text and image. Therefore, the correct choice is NO.2.

D Baselines

We compare six text modality approaches, three image modality approaches, four classical multimodal approaches, and one approach specifically designed to solve a multimodal Chinese idiom reading comprehension task. These methods are described in detail below.

- **BERT-WWM** [5]: An upgraded version of the BERT model that uses whole word masking.
- **RoBERTa** [5]: An improved BERT employs dynamic masking, more pre-training data, and extended training time.
- **MacBERT** [5]: An improved version of BERT, specifically designed for Chinese, introduces the pre-training task of MLM as correction.

- **PRIEM** [33]: Fusing the definitions of idioms through the prompt method and then using orthogonal projection to distinguish idioms' representations.
- **RISCF** [46]: The accuracy of idiom completion tests has been enhanced through semantic contrast learning and an anti-interference cross-attention module.
- **MSCLM** [45]: The model addresses the issues of metaphorical inconsistency and contextual inconsistency in idioms. By using metaphorical contrastive learning and multi-semantic cross-attention modules.
- **VGG-16** [35]: Enhancing model representation by stacking multiple smaller convolutional and pooling layers.
- **CLIP** [32]: A multimodal pretraining model designed to combine text and image information for robust cross-modal understanding and generation.
- **DeiT** [40]: Utilizing distillation to train small models by transferring knowledge from large pre-trained models, reducing model complexity and computational cost.
- **MSCA** [18]: A multimodal stack cross-attention network for better alignment and fusion of multimodal token-level features for the multimodal fake news detection task.
- **Multi-view CLIP** [31]: A framework that is capable of leveraging multi-grained cues from multiple perspectives for the multimodal sarcasm detection task.
- **DivE** [22]: A cross-modal retrieval approach utilizing smoothed chamfer similarity and using ensemble prediction modules.
- **CLTL** [44]: A method that won first place for hate speech target detection in the Multimodal Hate Speech Event Detection Challenge 2024.
- **DCIGN** [42]: The first method for multimodal Chinese idiom reading comprehension, which provides a dual contrastive learning loss function to align text and image features from coarse to fine.

E Algorithm Flowchart

In order to demonstrate the overall architecture of the MePe framework, clarify the relationship between modules, and illustrate the direction of data flow more intuitively, we provide an algorithmic flowchart for MePe, as shown in Algorithm 1.

F Limitations

In this paper, we rethink the multimodal Chinese idiom reading comprehension task by integrating a literal metaphorical semantic graph and a mixture of idiom experts. We validate the effectiveness and superiority of the proposed method through extensive experiments on the multimodal Chinese idiom reading comprehension dataset. However, there are still some shortcomings. (1) Due to the scarcity of the dataset, our proposed method only performs well on multimodal Chinese idiom reading comprehension, and its performance on other multimodal Chinese language forms, such as Chinese two-part allegorical sayings, literacy, and poetry, needs to be verified. (2) Our proposed method only considers the relationship between literal and metaphorical semantics among candidate Chinese idioms. Future work can explore the metaphorical relationships in images corresponding to Chinese idioms to further enhance the model's understanding of Chinese idioms.

Algorithm 1 MePe Algorithm

Input: Given an image-text pair P and I , A combination containing candidate idioms $Idiom_k$, and the corresponding label Y .

Output: Predicted probability $\hat{Y}$, optimized parameters Θ .

```

1: for Text and Image for each sample do
2:    $T_o$  and  $V_o$  are obtained through Eq. (1).
3:    $\tilde{T}_o$  and  $\tilde{V}_o$  is obtained through Eq. (2).
4:    $\mathcal{F}_M$  is obtained through Eq. (3).
5:   for  $\tilde{T}_o, \tilde{V}_o$  do
6:      $\mathcal{L}_{ccl}$  and  $\mathcal{L}_{fcl}$  are calculated through Eq. (4).
7:   end for
8:    $\mathcal{L}_{del}$  is obtained through Eq. (5).
9: end for
10: for Candidate idioms for each sample do
11:    $C$  is obtained through Eq. (6).
12:    $L$  and  $M$  are obtained through Eq. (7).
13:    $\mathcal{H}$  is obtained through Eq. (8). Select  $L_n$  and  $M_n$ .
14:    $\mathcal{A}_{ij}^{(0)}$  is constructed through Eq. (9).
15:    $e_{t,h,i,j}^{(r)}$  is calculated through Eq. (11).
16:    $\tilde{\mathcal{A}}_{t,i,j}^{(r)}$  is calculated through Eq. (13).
17:    $U_{lit}$  and  $U_{met}$  are obtained through Eq. (16).
18:    $\mathcal{F}_C$  is calculated through Eq. (17).
19:    $\mathcal{L}_{mmd}$  is calculated through Eq. (20).
20: end for
21: for  $\mathcal{F}_M, \mathcal{F}_C$  do
22:    $L_o$  is obtained through Eq. (21-22).
23:    $M_o$  is obtained through Eq. (23-24).
24:    $\mathcal{F}_{final}$  is obtained through Eq. (25).
25:    $\mathcal{L}_{lbl}$  is calculated through Eq. (26).
26: end for
27:  $P(Idiom_k|P, I)$  is calculated through Eq. (27).
28: Compute  $\mathcal{L}_{ce}$  through Eq. (28).
29: Compute  $\mathcal{L}$  through Eq. (29).
30: Apply  $\mathcal{L}$  to optimize parameters  $\Theta$ .
```

G Theory of Metaphor

Johnson and Lakoff [21] proposed theory of conceptual metaphor posits that metaphor is not merely a linguistic embellishment, but rather a conceptual system grounded in perceptual experience. This implies that metaphor recognition requires attention to conceptual mapping mechanisms, not just surface linguistic features. It argues that metaphor is fundamentally conceptual in nature, not merely a linguistic embellishment. We understand and construct abstract concepts through systematic mapping from a concrete, familiar "source domain" (e.g., "journey") to an abstract, unfamiliar "target domain" (e.g., "love"). For instance, "Our relationship has hit a dead end" maps obstacles in a 'journey' to difficulties in a "relationship." Gibbs et al. [14] provided further details, noting that idiom comprehension involves both non-combinatorial and combinatorial models, directly reflecting the complexity of metaphor processing. Fuyama [12] demonstrated through a quantum cognition model that literal and figurative meanings can form superposition states

during metaphor comprehension. This unique interaction pattern aids in distinguishing metaphorical from literal semantics.

H MLLMs

The introduction of six MLLMs is as follows.

- GLM4V³[15]: Developed by ZhipuAI, this is a multimodal large language model that centers on visual understanding. It is excellent at connecting visual inputs with textual reasoning to handle various tasks.
- CogVLM2⁴: Developed by ZhipuAI, a Chinese multimodal large language model that specializes in integrating Chinese text and visual data for efficient cross-modal understanding.
- InternVL2-Pro⁵: Developed by OpenGVLab, it draws on cutting-edge visual-language alignment to deal with complex scene analysis as well as content generation.
- Qwen1.5⁶ [38]: Developed by Aliyun, it offers support for multilingual interactions and stands out in multimodal scenarios that are code-related and data-driven.
- GPT-4o: Developed by OpenAI, it excels at processing text, images, and audio at the same time, thereby enabling real-time cross-modal reasoning.
- GPT-5: Developed by OpenAI, it is the first AI model to blend the reasoning abilities of the O-series models and the rapid response capacities of the GPT series models.

I Prompt Template

We present prompt templates for multimodal large language models, as shown below.

Prompt for MLLMs testing

You are an expert in Chinese idioms. According to the image and sentence provided, which four-letter idiom should be used to replace “idiom” in “sentence”? ‘Candidates’ answers are: “[candidate1, ..., candidate7]”. Please choose the correct answer:

对多语言大语言模型的提示

你是一个中文成语完形填空专家，根据提供的这一张图片[image]和一个句子[sentence]，句子中的“idiom”应该替换成哪个四字成语？候选答案有：[候选成语1, ..., 候选成语7]，请你选择正确的答案：

³<https://github.com/THUDM/GLM-4>

⁴<https://github.com/THUDM/CogVLM2>

⁵<https://github.com/OpenGVLab/InternVL>

⁶<https://tongyi.aliyun.com/qianwen/>